

radio circuitry. Carrier frequency of the module used here is 434MHz.

L293D. This is a dual H-Bridge high-current motor driver IC. It is used here since digital pins of Arduino cannot source enough current to drive the motors of the robotic car. H-bridges are also useful in controlling the direction of rotation of the motors.

Enable pins (pins 1 and 9) of the IC being active high are connected to 5V. Input pins IN1 through IN4 of L293D are connected to output pins D11 through D8 of HT12D, respectively. Output pins OUT1 through OUT4 of L293D IC are connected to motors M1 and M2, as shown in the circuit diagram of receiver side.

7805. This is a three-terminal, 5V fixed-voltage regulator with output current up to 1A. Although designed primarily as a fixed-voltage regulator, it can be used with external components for adjustable voltage and current.

Circuit diagrams of transmitter and receiver sides are shown in Figs 6 and 7, respectively.

Software

Arduino IDE 1.6.5 is used for programming Arduino Uno. The GUI application program was developed in R2014a version of MATLAB. The procedure for installing Legacy MATLAB and Simulink Support for Arduino package was described in 'Logging Sensor Data In MS EXCEL Through MATLAB GUI' article published in June 2018 issue. The same is also available on <https://electronicsforu.com>

After setting up the path for the package, open the source code file for this project, titled `wireless_car.m`, from MATLAB. Edit the COM port (in the line, `a = arduino('COM8')`) with the port number in your PC where Arduino Uno board is installed.

After running the file, MATLAB will try to communicate with the board. After successful communication is established, you can wirelessly control the robot by pressing the appropriate pushbutton in the GUI.

A particular pushbutton of the GUI, when pressed, executes callback function corresponding to it in this MATLAB program, `wireless_car.m`. Within that function, instructions to set/reset digital I/O pins 9, 10, 11 and 12 of Arduino are executed.

Four digital I/O pins of Arduino are connected to data pins AD8 through AD11, respectively, of HT12E encoder. After a pin is set or reset, the encoder scans the status of the digital pins and transmits the digital pins' status wirelessly to receiver side using 434MHz wireless transmitter (TX1).

Receiver side receives the data, and HT12D decoder present on receiver side decodes it and sets its output pins D8, D9, D10 and D11, accordingly. These output pins of the decoder are, in turn, used to control output pins of L293D motor driver. Signal logic levels at different stages of the circuits for proper

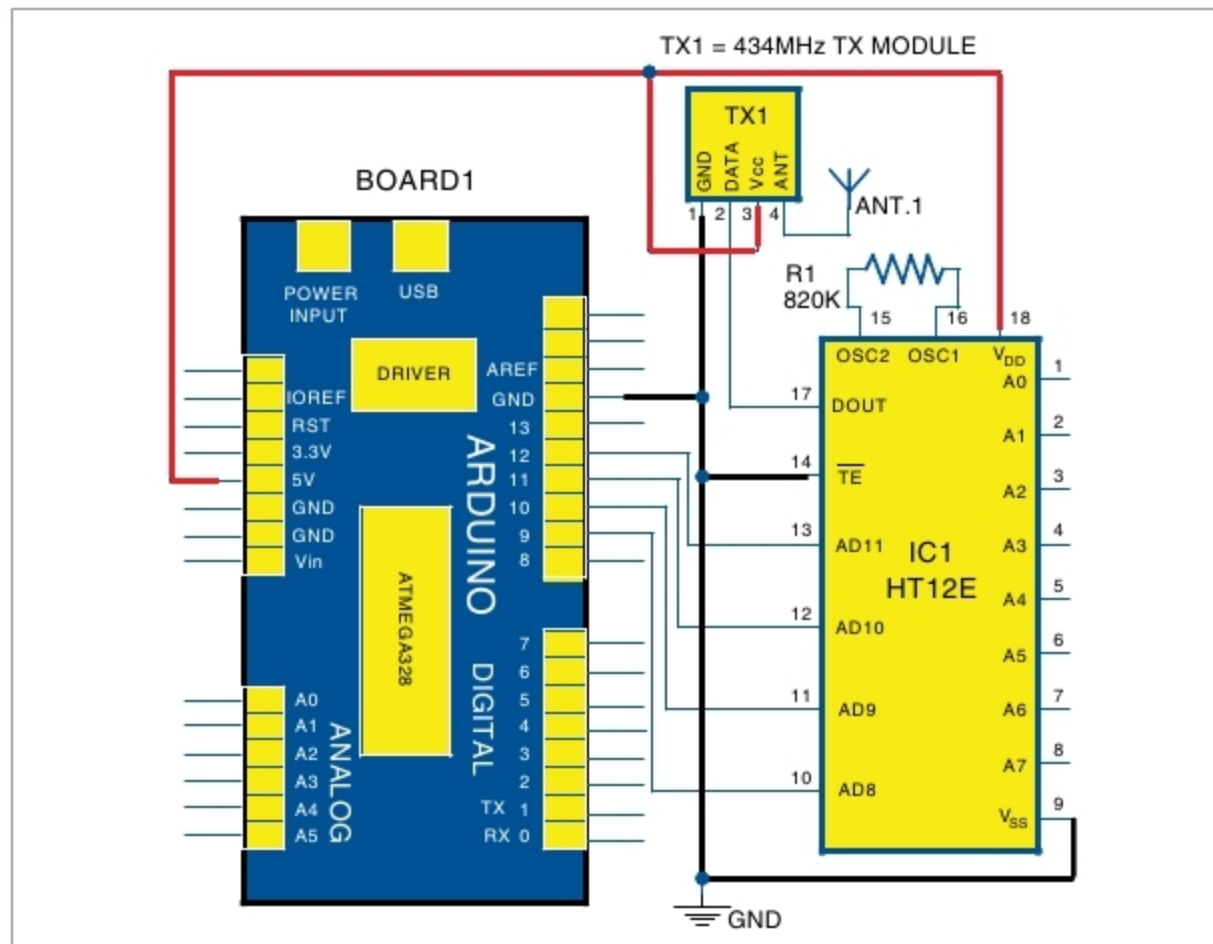


Fig. 6: Circuit diagram of transmitter side

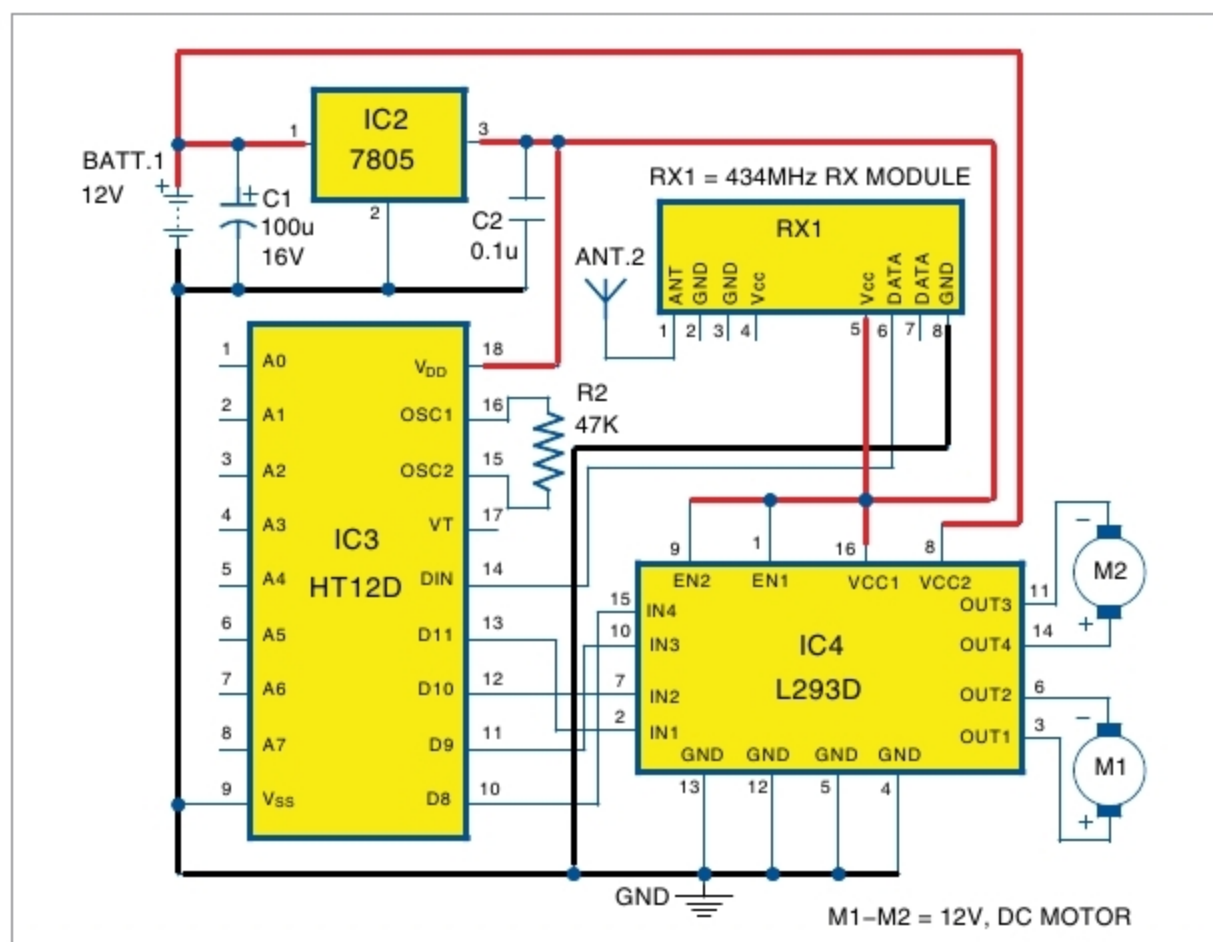


Fig. 7: Circuit diagram of receiver side